

Reflection

AI tools were most helpful during the development process by saving time when exploring the dataset, debugging code, and understanding the structure of the SAPIENT messages. They helped me understand the generated sensor data, design a simple anomaly detection method, and convert multiple sensor detections into a short situation summary for a human operator. The LLM was useful for turning raw sensor information into a more readable summary that is easier to understand.

However, AI suggestions were not always fully correct and needed to be checked against the original data. Some suggestions were too general and did not always consider the details of the dataset. For example, the LLM summary correctly identified altitude changes and confidence patterns, but it made the conclusion that every repeatedly detected object showed decreasing altitude. In reality, only some objects showed this pattern within the selected time period. The LLM also used the word "targets", although the data only contained object IDs and sensor measurements. This shows that LLM outputs should always be reviewed before being used for decision-making.

If this system were used in production, the biggest challenges would be reliability, false alarms, and missing context. The current anomaly detection method is simple and can identify unusual patterns, but it may also flag normal situations as anomalies. A real system would need more historical data, better tracking methods, domain knowledge, and validation from human operators. The LLM summary would also need strong connection to verified sensor data to reduce incorrect information. Overall, AI tools improved the development process, but human review is still important, especially for systems used in safety-critical situations.

Overall, AI tools helped throughout the project by improving data understanding, supporting the implementation of anomaly detection, and generating a useful situation summary. At the same time, the results showed that AI outputs need careful checking, and a production system would require stronger validation, better context handling, and human supervision.